Phaenomena

ΑΡΑΤΟΥ ΦΑΙΝΟΜΕΝΑ

Aratus

translated by Claude AI

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Phaenomena

Proem

1

From Zeus let us begin, whom we men never leave unspoken. All the streets are full of Zeus, and all the gathering-places of men; full of him is the sea, and the harbours; at every turn we all have need of Zeus. For we are of his stock as well; and he, gentle toward men, gives signs that are favourable, and rouses the peoples to work, putting them in mind of their living; he tells when the clod is best for oxen and for mattocks, he tells when the seasons are favourable both for trenching about young trees and for casting every kind of seed. For it was he himself who made the signs fast in heaven, setting the constellations apart, and he looked out for the year the stars that should give to men the most firmly appointed signs of the seasons, so that all things may grow without fail. That is why men always seek his favour first and last. Hail, Father, great wonder, great blessing to mankind, yourself and the earlier generation[1]. And hail, Muses, most gentle every one; and for me, as I pray to tell of the stars so far as is right, mark out the way for all my song.

The Northern Constellations

19

The stars, many as they are and scattered this way and that, are all alike drawn across heaven every day, continuously, for ever; but the axis shifts not even a little, but just as it is stays always fixed, and holds the earth in the midst, evenly balanced on every side, and drives heaven itself around. Two poles bound it at either end; but the one is not to be seen, while the other, facing it in the north, stands high

above Ocean. Two Bears, holding it between them, wheel round it together, and for that reason they are called the Wains[2].

28

Now they keep their heads always toward each other's loins, and always they travel shoulder-foremost, turned opposite ways upon their shoulders. If the tale is true, those Bears mounted into heaven from Crete by the will of great Zeus, because when he was then a child on fragrant Dicton, near the Idaean mountain, they laid him in a cave and nursed him for a year, while the Dictaean Curetes were deceiving Cronus. One of them men call by the name Cynosura, the other Helice. By Helice the Achaeans at sea take their mark for where to steer their ships, while the Phoenicians cross the sea relying on the other. Now Helice is clear and easy to make out, showing large from the first of the night; the other is slight, yet the better for sailors, since the whole of her turns in a smaller circuit; and by her the Sidonians sail straightest of all.

45

Between the two, like a branch broken from a river, winds a great wonder, the Dragon, bent round and about, immeasurable; and on either side of his coil the Bears are borne along, keeping clear of dark-blue Ocean. Toward the one he reaches out with the end of his tail, and the other he cuts off with his coil. The tip of his tail stops short beside the head of the Bear Helice, and Cynosura has her head within the coil; the coil winds right past her head and goes as far as her foot, and then turns back again and runs upward. Nor is it a single star shining alone upon that head of his, but two on the temples and two on the eyes; and one beneath them holds the extremity of the dread monster's jaw. His head is aslant, and looks altogether as if it were nodding toward the tip of Helice's tail; his mouth and the right side of his temple are in a very straight line with the end of her tail. That head, it may be, passes just where the limits of settings and of risings mingle with one another.

63

Right there rolls a figure like a man labouring. No one knows how to say plainly what it is, nor on what labour that man hangs, but they simply call him the Kneeler[3]. That figure, toiling on its knees, looks like a man sinking to a crouch; from both his shoulders his arms are lifted, and they stretch, one this way and one that, a full fathom's span; and he holds the tip of his right foot above the middle of the crooked Dragon's head. There too is that Crown which illustrious Dionysus set to be the sign of Ariadne when she was gone; it turns beneath the back of the wearied Figure.

74

The Crown comes near his back; but by the top of his head look for the head of the Serpent-Holder, and from it you may make out the shining Serpent-Holder himself, so bright do the shoulders appear that lie beneath his head; those would be visible even with the moon at mid-month. But his hands are not at all their equal, for faint is the gleam that runs over them on this side and on that. Yet even so they too can be seen, for they are not slight; and both are busy with the Serpent, which coils about the Serpent-Holder's middle. And he, steadfast and well planted, presses down with both his feet on the great beast, the Scorpion, standing upright on its eye and on its breast. But the Serpent writhes in his two hands, little of it in the right, but in the left much, rising high. And indeed the tip of its jaw leans close to the Crown, and beneath its coil look for the great Claws; but they are poor in light, not brilliant at all.

91

Behind Helice, like a man driving, comes Arctophylax, whom men also name Boötes, because he seems to be laying hands on the Bear that is a Wain. All of him is very clear to see; and beneath his belt wheels Arcturus himself, a star that stands out plainly from the rest.

96

Under the two feet of Boötes look for the Maiden; in her hands she carries the gleaming Ear of Corn[4]. Whether she is the child of Astraeus, who they say was the ancient father of the stars, or of some

other, may she go her way in peace. But another tale runs among men: how once, they say, she lived upon the earth and came face to face with mankind, and never spurned the tribes of men or of women of old, but sat down in their midst, immortal though she was. And they called her Justice; and gathering the elders together, in the market-place perhaps or in the broad street, she would sing, urging on them judgements for the people's good. Not yet in that time did they know of baneful strife, nor of dispute with all its blame, nor of the din of battle, but lived simply as they were; and the cruel sea lay far away, and ships did not yet bring livelihood from afar, but oxen and ploughs and she herself, the mistress of the peoples, Justice, giver of just things, furnished all things in countless plenty. So it was as long as the earth still nourished the golden race.

115

But with the silver race she kept company little, and no longer with full readiness, longing for the ways of the peoples of old. Yet even so she was still there in the time of that silver race. She would come down from the echoing mountains toward evening, alone, and spoke with no one in gentle words; but when she had filled the great hills with people, then she would threaten them, laying hold of their wickedness, and said she would no longer come before their eyes when they called: "What a race your golden fathers left behind them, worse than themselves! And you will breed children more evil still. And then, surely, there will be wars, and then too hostile bloodshed among men, and grievous pain will be laid upon them." So saying she made for the mountains, and left the people all still gazing after her.

129

But when those too were dead, and these were born, the bronze generation, men more deadly than those before them, who first forged the evil-working blade of the roadside robber, and first fed on the oxen of the plough, then Justice, hating the race of those men, flew up to heaven; and she settled in that place where by night she

still shows herself to mankind as the Maiden, close to conspicuous Boötes.

137

Above both her shoulders wheels a star on her right wing, and it is called in its turn the Vintage-Herald[5], as great in size and set with as much brightness as the one that shows beneath the tail of the Great Bear. For awesome is that Bear, and awesome are the stars near her; once you had seen them you would look for no further sign of those that, wheeling behind her, mark out her whole shape, so fair and so large a star travels before her feet: one before the feet beneath her shoulders, one before those that come down from her loins, and another beneath her hind knees. But all of them, single, one here and one there, are borne along in namelessness.

147

Beneath the Bear's head are the Twins, and beneath her middle is the Crab, and under her hind feet the Lion shines fair. There lie the sun's hottest summer paths; and the fields, it may be, are seen empty of corn-ears when the sun first comes together with the Lion. Then too the roaring Etesian winds fall in a mass upon the wide sea, and voyaging is no longer in season for oars. Broad ships would be my pleasure then, and let the helmsmen hold the steering-oars into the wind.

156

But if it is your mind to observe the Charioteer and the stars of the Charioteer, and word has reached you of the Goat herself and of the Kids, who have often looked upon men being scattered on the darkly heaving sea, you will find him, the whole of him, huge, lying over to the left of the Twins; and the top of Helice's head wheels opposite him. On his left shoulder is set the sacred Goat, who, as the tale goes, gave her breast to Zeus; the interpreters of Zeus call her the Olenian Goat. Now she is large and bright, but the Kids there shine faintly at the wrist of his hand.

167

By the feet of the Charioteer look for the horned Bull, crouching. The signs set for him are a very good likeness, so clearly is his head marked out; nor would anyone judge the head of the ox by any other sign, so well do the stars themselves, wheeling on either side, shape it. Their name too is much spoken, and the Hyades are by no means unheard of. They are strewn over the whole brow of the Bull; and the tip of his left horn and the right foot of the Charioteer who lies beside him are held by a single star, and they are borne along fastened together; but always the Bull is ahead of the Charioteer in going down to the other side, though he came up in his company.

179

Nor shall the suffering family of Cepheus, of the line of Iasus, be left lying just so, unspoken; but their name too has come into heaven, since they were near to Zeus. Cepheus himself, placed behind the Bear Cynosura, is like a man stretching out both his arms; the line that runs from the end of her tail to each of his feet is equal to that which stretches from foot to foot. And you would have to look only a little way aside from his belt if you were seeking the first bend of the great Dragon.

188

Before him rolls ill-fated Cassiopeia, not very large when she shines on a night of full moon; for not many are the stars that brighten her, and they alternate, ranging in rows to outline the whole of her distinctly. Like the key with which men, attacking a double door fastened on the inside, knock back the bolts, such is the look of the stars that singly underlie her. And she just stretches out a fathom's reach from her small shoulders; you would say she was grieving over her child.

197

For there too wheels that dread image of Andromeda, set in splendour beneath her mother. I do not think you will have to scan the night all round to see her at once, such is her head, such the shoulders on either side of her, and the tips of her feet, and all her girdle. Yet even there she is stretched out with arms flung wide, and bonds are laid on

her even in heaven; and those hands of hers are held up there, outspread, through all the days.

205

But against her head is driven the monstrous Horse with the underside of his belly; and a star in common shines on his navel and on the crown of her head. Three other stars besides, on the flanks and shoulders of the Horse, mark off equal distances as they stand apart, fair and large; but his head is in no way like them, nor his neck, long though it is. Yet the last star, on his blazing jaw, might vie even with the former four, which enclose him and are very conspicuous. Nor is he four-footed: from the tip of his navel the sacred Horse wheels round half-finished at the middle. He it was, men say, who brought down from lofty Helicon the fair water of the nourishing Horse's Spring. For the summit of Helicon was not yet streaming with springs, but the Horse struck it, and from that very spot the water gushed out in a flood at the blow of his forefoot; and the herdsmen were the first to give that draught its name, the Horse's Spring. But that water trickles from the rock, and you will never see it far from the men of Thespiae; whereas the Horse wheels in the house of Zeus[6], and there you may gaze on him.

225

There too are the swiftest paths of the Ram, who, though he is driven round the longest circuit, runs not a whit more slowly than the Bear Cynosura: himself faint and starless, as if looked at by moonlight, yet you may still judge his place by the girdle of Andromeda, for he is set a little beneath her. He treads his way through the middle of the great heaven, where the tips of the Claws and the belt of Orion wheel round.

233

There is yet another sign fashioned close by, below Andromeda: the Triangle, measured out on three sides, with two of them looking as though they were equal; the third is not so long, but very easy to find, for it is well starred beyond many. A little to the south of these are the stars of the Ram.

239

Still further on, and still in the approaches of the south, are the Fishes. But always one is farther forward than the other, and hears more of Boreas when he comes freshly down. From both of them there stretch, as it were, bonds from their tails, coming on either side in unbroken line to a single point. And one star holds them, fair and large, which men call the Knot beneath the Tails[7]. Let Andromeda's left shoulder be your sign for the more northerly Fish, for it is very close to it.

248

And both her feet may point out her bridegroom Perseus, for they always travel upon his shoulders. He moves in the north, taller than the others. His right hand is stretched toward the seat of his bride's mother's chair; and as though chasing what lies before his feet he lengthens his strides, covered with dust, in father Zeus.

254

Near his left knee the Pleiades all travel in a cluster. The space that holds them all is not very large, and they are themselves dim to observe. Seven-pathed they are called among men, though only six are visible to the eyes. Yet no star at all has perished unremarked out of Zeus in all the time of which we have heard since our race began, but so, simply, the tale is told. And those seven are called by name Alcyone and Merope and Celaeno and Electra and Sterope and Taygete and lady Maia. Small alike they are and without brilliance, yet renowned they wheel at morning and at evening, and Zeus is the cause, who bade them give the sign of summer and of winter beginning, and of the ploughing as it comes on.

268

And there is the Tortoise, a small one. While he was still beside his cradle Hermes bored it through, and said that it should be called the Lyre. And he brought it into heaven and set it down in front of the unknown Figure. The Figure, crouched upon its legs, comes near it with the left knee, while the top of the Bird's head wheels on the

other side; and the Lyre is fixed midway between the Bird's head and the knee.

275

For indeed a shimmering Bird runs along with Zeus. Hazy he is in part, but other parts upon him bristle with stars, not very large, yet not dim either. Like a bird in fair-weather flight he is carried before the wind toward the other side, stretching the right tip of his wing toward the right hand of Cepheus, while beside his left wing leans the prancing of the Horse.

282

And as the Horse bounds on, the two Fishes range on either side of him; and beside his head the right hand of the Water-Pourer is stretched out. The Water-Pourer rises after Aegoceros; but Aegoceros lies before him and lower down, where the might of the sun turns back.

287

In that month may you not be washed round by the sea, having ventured on the open deep. Neither by day could you make much way, since the days are then at their swiftest; nor, when you are in terror by night, would the dawn come near, however loudly you cried out. That is when the grievous blasts of Notus come bursting down, when the sun meets with Aegoceros; and then the cold from Zeus is crueller to the benumbed sailor. Yet for all that, the sea now heaves dark beneath the keels all the year long; and like diving shearwaters we often sit peering round from the ships over the deep, turned toward the beaches; but they are still far off, washed by the waves, and a little timber keeps Hades away.

300

And in the month before as well, when you have suffered much from the sea, when the sun burns the Bow and the Drawer of the Bow, put in to land at evening and trust the night no longer. Let your sign of that season and that month be the Scorpion rising at the last of the night. For the Archer draws his great bow close by its sting; and the

Scorpion, as it rises, stands a little ahead of him, and he comes up straight after. Then too the head of Cynosura runs very high at the end of night, and Orion sets whole before the dawn, and Cepheus from his hand to his loins.

311

Further on there is another Arrow shot, by itself, without a bow; and beside it the Bird is outspread, nearer to the north. And close to it another bird is blown along, not so great in size, yet harsh when it comes up from the sea as the night departs; and they call it the Eagle. The Dolphin, not very large, runs above Aegoceros, hazy in the middle; but about him lie four jewels, set side by side, two and two.

The Southern Constellations

319

Now these are spread between the north and the wandering of the sun; but lower down others rise, many of them, between the south and the sun's path. Aslant beneath the cut of the Bull[8] lies Orion himself. Let no one who passes him by on a clear night, when he is ranged on high, trust that in looking up at heaven he will behold other things more splendid.

326

Such too is his guard, seen beneath his high-lifted back, standing upon both feet: the Dog, spangled, yet not bright in every part; along the belly itself he wheels round dark, but the tip of his dread jaw is set with a star that above all others blazes searingly; and men call it Sirius. When he rises with the sun, the orchards can no longer deceive him with their feeble show of leaves. For easily, darting keenly through their rows, he sorts them: some he strengthens, and of others he destroys all the bark. Of him we hear at his setting too; but the other stars lie round about, slighter, to mark his limbs.

338

Under both feet of Orion the Hare is hunted unceasingly, every day; and Sirius comes on always behind him like one in pursuit, and rises after him, and watches him as he goes down.

342

By the tail of the great Dog Argo is hauled along stern-first; for hers is no course of the customary kind, but she travels turned backward, as real ships do when the sailors have already swung the stern about on entering the anchorage: at once every man backs the ship, and surging sternward she takes hold of the land. So this Argo of Jason is hauled along stern-first. Part of her, from the prow right up to the mast, moves hazy and starless, but the rest is all bright. And her steering-oar, let down, is fixed beneath the hind feet of the Dog as he goes before.

353

And Andromeda, though she cowers at no small distance, is hard pressed by the great Sea-Monster as it comes on. For she moves along lying beneath the blast of Thracian Boreas, while Notus drives against her the hateful monster, under the Ram and both the Fishes, set a little above the starry River. For that too travels beneath the feet of the gods, the lone remnant of Eridanus, river of many tears. It stretches beneath the left foot of Orion; and the bonds of the tails, by which the Fishes are held at their ends, both come together as they descend from the tails; and behind the Sea-Monster's crest they travel mingled, driving on to one point, and they end in a single star of the Sea-Monster, which lies on the first of its spines.

367

Stars of small measure and endowed with small brightness wheel midway between the steering-oar and the Sea-Monster, cowering beneath the flanks of the grey Hare, nameless; for they are not set in the likeness of the limbs of a fashioned figure, like the many that pass in ordered ranks along the same paths as the years fulfil themselves: those that some man of the generations now gone marked, and devised how to call them all by name, shaping them in groups. For he

could not have told the name of all of them taken singly, nor learned them. For they are many on every side, and of many the sizes and the colour are alike, and all of them go circling round. Therefore he resolved to make the stars into companies, so that, lying one beside another in order, they might signify shapes. And at once the constellations got their names, and no longer now does a star rise to our wonder. But those appear fitted into clear figures, while the stars beneath the hunted Hare are all very hazy and move without a name.

386

Below Aegoceros, under the blasts of Notus, a Fish hangs poised, turned toward the Sea-Monster, alone and apart from the former ones; and they call him the Southern Fish. Other stars, lying scattered beneath the Water-Pourer, hover between the Sea-Monster of the sky and the Fish, faint and nameless; and near them, from the right hand of the illustrious Water-Pourer, like a slight pouring of water sprinkled this way and that, others wheel, pale-gleaming and feeble. Among them move two stars that show more clearly, neither very far apart nor very near: one beneath both feet of the Water-Pourer, fair and large, the other beneath the tail of the dark Sea-Monster. All these together they call the Water. A few others besides, below the Archer, under his forefeet, wheel round, curved into a rounded ring[9].

402

Beneath the blazing sting of the great monster, the Scorpion, near the south, hangs the Altar. You will find it aloft for only a little time, for it rises over against Arcturus; and while for him, Arcturus, the paths run very high overhead, it goes the sooner down beneath the western sea. Yet around that Altar too ancient Night, weeping for the toil of men, has set a great sign of storm at sea. For ships that are scattered are not to her mind, and she shows forth other signs from other quarters in her pity for men beaten by many waves. Therefore, I beg you, do not pray at sea that that constellation should appear in mid-heaven with other clouds wrapped round it, itself unclouded and bright, yet pressed upon from higher up by a billowing cloud, such as are often

packed tight when the autumn wind drives them back. For often Night herself contrives this sign too for Notus, doing a kindness to toiling sailors.

420

If the sailors heed Night when she gives the timely signal, and quickly make all light and trim, at once their toil is easier; but if a dread squall of wind strikes the ship from on high, wholly unforeseen, and throws all the sails into confusion, sometimes they make their voyage altogether under the waves, and sometimes, if by their prayers they win Zeus to come to their side, and the might of the wind Boreas flashes out in lightning, after much hardship they yet look on one another again aboard the ship. At this sign fear Notus, until you see Boreas flash his lightning.

431

But if the Centaur's shoulder is as far from the western sea as from the eastern[10], and a thin haze wraps the Centaur himself, while behind him Night fashions the like signs upon the all-gleaming Altar, you must look out not for Notus but for the wind Eurus. You will find that constellation lying beneath two others: the parts of it that are like a man lie below the Scorpion, and the Claws hold the horse's hindquarters under them. He looks like one ever stretching his right hand toward the rounded Altar, and in it, fast held, another creature is tightly gripped, drawn out through his hand: the Beast, for so the men of old named it.

443

But there is yet another constellation that trails along from the far side: they call it Hydra, and like a living thing it winds on at full length. Its head comes beneath the middle of the Crab, its coil beneath the body of the Lion, and its tail hangs over the Centaur himself. On the middle of the coil lies the Bowl, and on the last of it the figure of a Raven that seems to peck at the coil. And there too, beneath the Twins, Procyon shines fair. These you may watch as the years go by,

returning in order each at its season; for all of them, just so, are fixed fast in heaven, the ornaments of passing night.

The Planets

454

But there are five other stars mingled among them, in no way like them, that whirl every way through the twelve figures. No longer could you judge where these lie by looking to other stars, since all are wanderers; long are the years of their circling, and far apart lie the signs of their coming together into one. Nor am I bold any longer when it comes to them: may I be equal to telling the circles of the unwandering stars and the signs in the sky.

The Circles of the Sky

462

These circles lie like turned rings, four of them, for which there would be most desire and most use in one who scans the measures of the years as they run out. Signs lie about them all, very plainly marked, many in number and all set close together on every side. The circles themselves are unwandering and all fitted to one another; but in measure two are matched against two.

469

If ever on a clear night, when heavenly Night displays all her splendid stars to men, and none goes its way enfeebled by the mid-month moon, but all shine piercing through the dark; if ever at such an hour wonder came over your mind as you observed heaven cloven all the way round by a broad circle, or someone else standing by you pointed out that wheel ringed with brightness, they call it the Milk. No other circle that wheels is like it in colour, but in measure two of the four are as great, while the other two turn far smaller than they.

480

Of these the one lies near where Boreas comes down. On it are carried both the heads of the Twins; on it lie the knees of the firm-set Charioteer; upon it are the left shin and left shoulder of Perseus; and it holds the middle of Andromeda's right arm above the elbow: her palm lies higher, nearer to the north, and her elbow leans toward the south. The hooves of the Horse, the neck of the Bird with the crown of his head, and the fair shoulders of the Serpent-Holder wheel as they are driven round the circle itself. The Maiden is borne a little farther south and does not touch it, but the Lion and the Crab do. Those two lie set one after the other, and the circle cuts the one beneath the breast and belly as far as the groin, and the other, the Crab, straight through beneath the shell, where you may best see him divided upright in two, so that his eyes go one on either side of the circle. If the circle is measured, as nearly as may be, into eight parts, five turn in the open sky above the earth and three on the far side; and in it is the turning of summer. This one is fixed in the north, about the Crab.

501

Another, in the south that faces it, cuts the middle of Aegoceros, the feet of the Water-Pourer and the tail of the Sea-Monster. On it is the Hare; of the Dog it takes no great share, only as much as he covers with his feet; on it is Argo, and the great back of the Centaur; on it the sting of the Scorpion; on it too the Bow of the illustrious Archer. This is the last that the sun comes to as he passes from the clear north toward the south, and there he makes his winter turning. Three parts of its eight wheel round on high, and five revolve sunk beneath the ground.

511

Midway between the two, as great as the circle of the grey Milk, a circle passes round beneath the earth like one that cuts in half; on it the days are made equal to the nights at both seasons, when summer wanes and again when spring begins. As its sign lie the Ram and the knees of the Bull: the Ram driven lengthwise along the circle, but of the Bull only so much of the legs as shows bent in a crouch. On it is the

belt of bright-beaming Orion and the bend of blazing Hydra; on it too the slight Bowl, on it the Raven, on it the stars, not very many, of the Claws; and on it the knees of the Serpent-Holder are carried. Nor indeed is it bereft of the Eagle, but close to it the great messenger of Zeus rides the wind. And along it the Horse's head and neck wheel round.

525

These, set upright and parallel, the axis holds cast about it, keeping them all at the middle; but the fourth is clamped aslant between both the tropics, which hold it on either side opposite each other, while the middle circle cuts it in the midst. Not otherwise would a man schooled in the handicraft of Athena join whirling wheels, such and so many, rolling them all round in the fashion of a sphere, than those in the sky, bound together by the slanting circle, speed from dawn to night through all the days.

534

Those rise and straightway set below, all side by side; and each of them has a single place of going down and of coming up, in order, on either side. But this one passes over as much of the water of Ocean as rolls, as nearly as may be, from the rising of Aegoceros to the rising of the Crab; as much as it covers in all at its rising, so much it covers on the other side at its setting. As long as is the ray cut off from the glance of the eye[11], six times that length would run beneath it; and each, measured equal, cuts off two constellations. They call it by name the circle of the Zodiac.

545

On it is the Crab, and after him the Lion, and beneath him the Maiden; after her the Claws and the Scorpion itself, and the Archer and Aegoceros, and after Aegoceros the Water-Pourer; beneath him the two Fishes are set in stars; after them the Ram, and after him the Bull and the Twins. Through all these twelve the sun travels as he leads on the whole year, and as he goes round this circle all the fruit-bearing seasons grow. As much of it as sets in hollow Ocean, so much is carried

above the earth; every night six twelfths of the circle always set, and as many rise. Each night is always stretched to such a length as it takes half the circle to lift above the earth from the beginning of night.

Risings and Settings

559

It would be no thing to throw aside, for one watching for the day, to observe when each of those portions rises; for always with one of them the sun himself comes up. You may scan them best by looking to the portions themselves; but if they should be darkened by clouds, or rise hidden by a mountain, make for yourself sure signs of their coming on. Ocean himself would give you them on both his horns[12], in the many stars he wreathes about himself, whenever he brings each of those portions up from below.

569

Not the faintest are the stars that lie wheeling about Ocean on both sides when the Crab rises, some setting, others coming up from the other side. The Crown sets, and the Fish[13] sets as far as his spine. Half of the Crown you may see aloft, and half is already cast down by the world's edges as it descends. But he who is turned backward has not yet gone down in the parts about his lower belly, while his upper parts are carried in night. The Crab also brings down the toiling Serpent-Holder from the knees to the shoulders, and brings down the Serpent close to the neck. Nor would Arctophylax any longer be large on both sides, the lesser part in day and the greater already in night. For Ocean receives Boötes as he goes down along with four portions together; and when he has had his fill of light, he takes up more than half the passing night with his ox-loosing[14], at the season when he sets as the sun goes down. Those nights are even named after his late setting. So these set; but facing them, in no way mean, but shining fair in his belt and fair in both shoulders, Orion, trusting in the might of his sword and bringing the whole River, stretches along the other horn.

590

As the Lion comes, all those go wholly down that were setting with the Crab, and the Eagle too. But he who sits on his knees is already down in the rest, yet his left knee and foot he does not yet turn beneath surging Ocean. And there rise the head of Hydra and the pale-gleaming Hare, and Procyon and the forefeet of the blazing Dog.

596

Not few indeed are the stars the Maiden casts beneath the depths of the earth as she rises. Then the Cyllenian Lyre and the Dolphin set, and the well-wrought Arrow; with them the foremost feathers of the Bird, right up to the tail itself, and the outstretched reaches of the River go into shadow; the Horse's head sets, and his neck sets too. Hydra rises further, as far as the Bowl itself, and the Dog, coming before, takes up his other feet, dragging behind him the stern of many-starred Argo. And Argo runs above the earth, parted in two at the very mast, when the Maiden has just come whole from the far side.

607

Nor would the Claws at their coming on, though they shine but faintly, pass unmarked, since Boötes, a great sign, rises all at once, set with Arcturus. Argo will by now be wholly aloft; but Hydra, for it is poured across heaven in vast length, would still want its tail. The Claws bring up only the right leg, as far as the knee itself, of him who is ever on his knees, ever crouched beside the Lyre: that one, whoever he is, unknown among the figures beneath heaven, whom we often watch in the selfsame night both setting and rising from the other side. Of him, then, the leg alone shows with both the Claws; he himself, still turned head-downward on the other side, waits for the rising Scorpion and the Drawer of the Bow. For they bring him: the one his middle and all the rest, while the Bow brings up his left hand together with his head. So he is carried up limb by limb, all in three parts; and half the Crown and the very tip of the Centaur's tail the Claws bring as they still rise.

627

At that time the Horse sets, following his vanished head, and the tail's end of the Bird that went before is drawn down after. Andromeda's head sets; and against her misty Notus brings on the great terror of the Sea-Monster, while facing it Cepheus himself from the north signals with his great hand upraised. The monster, turned toward its crest, sets as far as the crest itself, and Cepheus with head and hand and shoulder.

634

The bends of the River would plunge into fair-flowing Ocean as soon as the Scorpion comes on, which at its coming also puts great Orion to flight. May Artemis be gracious! It is a tale of the men of old, who said that mighty Orion laid hold of her by the robe, when in Chios he was striking down all the wild beasts with his stout club, striving to win for that Oenopion a gift of the chase. But she at once set upon him another beast, tearing apart the hills in the midst of the island on either side: a Scorpion, which stung him and killed him, great as he was, showing itself the greater, because he had vexed Artemis herself. That is why, they say, when the Scorpion comes from the far side, Orion flees round the ends of the earth.

647

Nor indeed are the remaining parts of Andromeda and the Sea-Monster unaware of the Scorpion while it is still rising, but they too flee in full haste. Cepheus then grazes the earth with his belt, dipping in Ocean all the parts toward his head, while the rest it is not lawful to dip, for the Bears themselves forbid it: his feet and knees and loins. And she too, hapless Cassiepeia, hurries after the image of her child. No longer do her parts show in due order from the chair, her feet and knees above, but she sets headfirst like a diver, parted at the knees[15], since she was not to set herself equal to Doris and Panope without great penalty.

659

So she is borne toward the other side; and heaven brings round others from below in answer, the second curve of the Crown and the end of

Hydra, and it brings besides the Centaur's body and head and the Beast which the Centaur holds in his right hand. But the forefeet of the horseman Centaur wait there for the Bow to come on. With the coming Bow rise the coil of the Serpent and the body of the Serpent-Holder; their heads the Scorpion itself brings as it rises, and it leads up the very hands of the Serpent-Holder and the foremost bend of the many-starred Serpent.

669

As for the Kneeler, since he always rises turned about, the rest of him then comes forth from the far side, his legs and belt and all his breast and his shoulder with the right hand; but his head with the other hand comes up with the Bow and the rising Archer. Together with them the Lyre of Hermes, and Cepheus as far as the breast, ride up from the eastern Ocean, at the hour when all the flashings of the great Dog set, and all of Orion goes down, and all of the Hare, pursued in a chase that has no end. But the Kids of the Charioteer and the Olenian Goat do not depart at once; they shine along his great arm, and are marked off from his other limbs for stirring up storms when they come together with the sun.

683

But some parts of the Charioteer, the head and the other hand and the loins, Aegoceros brings down as he rises; and all the lowest parts go down with the Archer himself. Neither Perseus nor the tip of the stern-post of many-starred Argo remains: Perseus sets save for his right knee and foot, and of the stern as much as reaches to the curve; she herself goes down as Aegoceros rises, at the hour when Procyon also sets and others come up, the Bird and the Eagle and the stars of the winged Arrow and the holy seat of the southern Altar.

693

When the Water-Pourer is newly rising, the Horse rolls up with feet and head; and over against the Horse starry Night drags the Centaur down by the tail. But she cannot find room for his head and his broad shoulders together with the breastplate itself. Yet she brings down the

neck-coil of fiery Hydra and all its brow. Much of Hydra still remains behind; but that too, with the Centaur himself, Night takes in all together when the Fishes rise. With the Fishes comes the Fish that lies beneath dark Aegoceros himself, not wholly, for a little of him waits for another twelfth.

704

So too the toil-worn hands and knees and shoulders of Andromeda are all stretched out in two parts, some before and others behind, when the two Fishes are newly come forth from Ocean. The parts by her right hand the Fishes themselves draw after them, and those on the left the Ram hauls from below as he comes up. And as he wheels round, you may see the Altar in the west, and on the other side as much of rising Perseus as his head and shoulders. His belt itself might well be disputed, whether it shows as the Ram is ending or at the Bull, with whom he rolls up in his entirety. Nor is the Charioteer left behind as the Bull rises, for he travels fitted close to him; yet he does not rise entire with that portion, but the Twins bring him up whole. The Kids, though, and the sole of his left foot, with the Goat herself, are carried along with the Bull, when the crest and tail of the Sea-Monster of the sky rise from the far side. And Arctophylax is by then already setting with the first of those portions, the four that bring him down, all but his left hand, which rises beneath the Great Bear herself.

724

Let both feet of the Serpent-Holder as he sets, right up to the knees, be made a sign for the Twins rising from the other side. Then no part of the Sea-Monster trails any longer on either side, but you will now see it whole. Now too a sailor on the clear open sea may watch the first bend of the River coming up out of the sea, as he waits for Orion himself, to see whether he may announce to him some sign of the measure of the night or of his voyage. For on every side the gods tell these many things to men.

Weather Signs

733

Do you not see? When the moon with slender horns shows from the west, she teaches of a month that is growing; when first a beam is shed from her, enough to cast a shadow, she is going on to the fourth day; eight days at her halving; and the mid-month with her full face. Ever tilting her brows now one way and now another, she tells which dawn of the month is wheeling round.

740

The ends of the nights, indeed, those twelve portions are sufficient to declare. But what bears on the great year, the season to plough the fallows and the season to plant, is all already revealed by Zeus and set on every side. And many a man aboard ship too has taken note of the much-surging storm, mindful either of dread Arcturus or of some other stars that are drawn from Ocean in the morning twilight, and those while it is still the first of night. For truly the sun passes them all in the course of the year as he drives his great furrow, and he will draw near now to one and now to another, at one time as he rises and at another as he sets; and a different star looks upon a different dawn.

752

You too know these things, for by now the nineteen cycles[16] of the shining sun are sung in concert, and all the stars that Night wheels round from the belt to the last of Orion and Orion's bold Dog, and the stars which, seen in Poseidon's realm or in that of Zeus himself, give men firmly appointed signs.

758

Therefore take pains over these things; and if ever you trust yourself to a ship, let it be your care to find out all the signs that anywhere lie appointed for stormy winds or for a tempest of the sea. The labour is small, but beyond counting is the profit that soon comes of heedfulness to the man who is always on his guard. First he is himself the safer, and then he does another man good by his warning, when a

storm has risen close at hand. For often a man, even under a calm night, makes his ship snug in fear of the sea at dawn; sometimes the trouble comes running on the third day, sometimes on the fifth, and sometimes it arrives unheralded. For we men do not yet know all things from Zeus, but much is still hidden, of which Zeus, if he will, shall give us hereafter as well; for he openly furthers the race of men, appearing on every side and everywhere showing his signs.

773

Some things the moon will tell you, perhaps when she is halved on either side of the full, or again when she is at the full; others the sun at his rising, and again when he gives his warning on the edge of night; and from other things too you will have other signs to make your own, concerning both night and day.

The Moon

778

Look first at the moon, at her horns on either side. For evening paints her at one time with one radiance and at another with another, and at different times different shapes horn the moon when she is just beginning to grow, some on the third day and some on the fourth; from these you may learn about the month that has set in. If she is slender and clear about the third day, she will be a sign of fair weather; if slender and very red, of wind; but if she is thicker and her horns are blunted, and her fourth-day light following the third is a feeble one, she is being blunted either by Notus or by rain that is near.

788

If, as she brings on the third day, she neither nods forward from both her horns nor shines leaning back, but the horns curve round upright on either side, westerly winds will sweep after that night. And if she brings in the fourth day too upright in the same way, surely she will be teaching that a storm is gathering. If the upper of her horns nods well forward, expect Boreas; when it lies back, Notus. But when on her third day a whole circle wheels about her, reddening all round, then

she will be stormy indeed; and for a greater storm she will glow a more fiery red.

799

Look to her at the full and at both her halvings, when she is growing and when she goes back again to a horn, and judge of each month by her colour. When she is clear all over you may judge that it will be very fair; when she is red all over, look for the paths of the wind; when she is dark in one place and another, look for rain. The signs are not all appointed for every day; but those that come on the third day and the fourth hold until the moon is halved, those of the half moon give their signs right up to the mid-month itself, and those of the mid-month in turn to the waning half; next upon that follows the fourth day of the departing month, and upon that the third of the month coming in.

811

If haloes encircle her all round, whether three or two lie about her or only one, with the single one expect wind or calm, wind if it is broken and calm if it fades away; two running round the moon would go with a storm; and a halo of three coils would bring a greater storm, and greater if it is dark, and greater still if it is broken. These, then, are the things you may learn for the month from the moon.

The Sun

819

Give heed also to the sun at either end of his going. In the sun are set signs still more to be trusted, both when he sets and when he comes up from the far side. Let his disc not be mottled when he first strikes the ploughlands, at a time when you have need of a fair day, nor let it bear any mark, but show plain all over. And if the hour of ox-loosing finds him clear in the same way, and he sets cloudless toward evening with a soft radiance, he will still bring fair weather at the coming dawn. But not when he rises seeming hollow, nor when his rays part and strike some toward the south and some toward the north, while

the middle of him shines bright; then, it may be, he is passing through rain or wind.

832

Look, if the sun's beams will give way to you, at the sun himself, for the watch kept on him is the best of all, to see whether any flush runs over him, as often he reddens in one place and another when clouds are trailing, or whether he is dark anywhere; and let the dark be your sign of rain to come, and all the reds of wind. But if he should be touched with both at once, he will both bring rain and strain along under wind. If at his rising or again at his setting his rays come together and throng about one point, or if at any time he goes hemmed in by clouds from night to dawn or from dawn to night, those days will run past with rain coming down.

845

Nor, when a little cloud rises before him and he himself lifts after it in want of his rays, be unmindful of rain. But when his disc broadens large about him, as though it were melting, at his first coming up, and then shrinks back to a smaller, he will run his course in fair weather; so too if in the winter season he pales as he goes down. But when rain has fallen by day, examine the clouds afterwards, turning toward the sun as he sets. If a cloud that seems to be blackening overshadows the sun, and the rays about it on this side and on that are parted in two as they wheel between, surely you will still be in need of shelter until the dawn.

858

But if he dips cloudless into the western stream, and as he goes down and when he is already gone the clouds stand near him red, you need not greatly tremble for rain on the morrow or in the night; but you must fear it when the sun's rays of a sudden stretch out from heaven like rays that are wasting away, in the way they are dimmed when the moon overshadows them, standing in a straight line between the earth and the sun. Nor, when he is slow to show himself before dawn and reddish clouds appear in one place and another, do the fields go

unwatered on that day. Nor likewise, when he is still on the far side and the rays stretched out before him show shadowed before dawn, forget that rain or wind will be coming down.

872

If those rays are borne along in a deeper gloom, they will be signs rather of rain; but if only a slight dusk is spread about the rays, such as soft clouds mostly carry, surely they are dusked over for a coming wind. Nor are dark haloes close about the sun signs of fair weather; the closer they are and the more unrelenting their blackness, the stormier; and two would be harsher still.

880

Look, at his rising or again at his setting, whether any of those clouds that are called parhelia[17] redden on the south or on the north or on both sides, and do not keep this watch feebly. For when those clouds hold the sun between them on both sides at once, close to Ocean, there is no putting off the storm that comes from Zeus. But if one alone glows crimson, on the north, it will bring the blasts of Boreas, and the southern one those of Notus; or perhaps drops of rain come running on. Trust these signs the more when they are in the west, for from the west their signalling is alike steadfast always.

The Manger

892

Look also at the Manger. Like a little mist it leads on in the north beneath the Crab; and about it two faintly shining stars are borne, neither very far apart nor very near, but as much, to the eye's best guess, as a short cubit, one on the north, while the other leans toward the south. These are called the Asses[18], and between them is the Manger. It will even, of a sudden, when Zeus is fair on every side, vanish altogether, while the stars that go on either side of it look close to one another: by no small storm are the fields then flooded. If it darkens, while both the stars remain like themselves, they will be giving a sign of rain. If the one to the north of the Manger shines

feebly, faintly misted, while the southern Ass is bright, expect the wind Notus; but for Boreas you must watch for the misted one and the shining one changed about.

Signs of Wind

909

Let a swelling sea too be your sign of wind, and beaches roaring far, and the headlands of the sea when in fair weather they fill with sound, and the high peaks of a mountain crying aloud. And when the heron in no due order comes from the sea to dry land, screaming loud and long, he is borne before a wind that is stirring over the sea. At times too the petrels, when they fly in fair weather, move in flocks to face the winds that are to come. Often wild ducks or sea-wheeling shearwaters beat their wings upon the land, or a cloud draws out at length along the peaks of a mountain. Before now thistledown too, the hoary age of the white thistle, has been a sign of wind, when in plenty it floats on the surface of the dumb sea, some ahead and some behind.

924

And in summer, from whatever quarter the thunders and the lightnings come, from that quarter look out for the wind coming on. And when through the black night stars shoot thick and fast, and the trails behind them whiten, expect a wind coming along the same road as they; but if others too shoot against them, and others from other quarters, then beware of winds of every kind, which are the most confused of all, and blow in a confusion beyond men's judging. But when it lightens from Eurus and from Notus, and now from Zephyrus and now from Boreas, then a seafaring man out on the open water is afraid that the sea may hold him on the one side and the water from Zeus on the other; for with rain so many lightnings travel round about.

Signs of Rain

938

Often, when rains are coming, clouds appear beforehand that look most like fleeces, or a double rainbow girdles the great heaven, or perhaps some star wears a darkening halo. Often the birds of the mere or of the sea plunge into the waters and wash themselves insatiably, or swallows dart long about a lake, striking with their bellies the water that is no more than ruffled, or those hapless tribes, a boon to the water-snakes, the fathers of the tadpoles, croak the louder from the water itself, or the solitary tree-frog[19] murmurs at daybreak, or perhaps the chattering crow, by a jutting shore, dips down upon the dry land[20] as the storm is coming, or perhaps bathes herself in a river from the head to the top of the shoulders, or even dives right under, or paces much to and fro by the water, hoarsely cawing.

954

And before now oxen, ahead of rain from the sky, have looked up to heaven and caught the scent of it from the ether; and ants have carried all their eggs up out of their hollow nest in haste; and millipedes have been seen in swarms crawling up walls, and wandering abroad those worms that men call the entrails of the black earth. And the tame fowl, the brood of the cock, have picked themselves well for lice and clucked aloud, with a sound like water dripping upon water. At times too the generations of ravens and the tribes of jackdaws have been a sign of rain coming from Zeus, appearing in flocks and crying with a voice like hawks. And ravens, it may be, have mimicked with their voice the drops of Zeus, when rain is on its way; or at times, croaking twice in a deep voice, they whirr long and loud, shaking out their thick feathers; and ducks that keep the house and jackdaws of the roof come to the eaves and flap their wings, or the heron speeds toward the wave with a piercing scream.

973

Let none of these be thrown aside when you are on your guard against rain; nor if flies bite more than before and lust after blood, or fungus gathers about the wick of a lamp on a dark night; nor if in the winter season the light of the lamps at one time rises in due order and at

another the flames dart off from them like light bubbles, nor if rays glitter upon the lamp itself, nor if, when summer is spread wide, the island birds pass over in close flights. Nor forget the pot or the fire-straddling tripod, when more sparks than usual are about it; nor when in the ashes of burning charcoal marks like millet seeds shine all about, but watch these too when you are looking out for rain.

Signs of Fair Weather

988

But if a misty cloud stretches along the base of a great mountain while the high peaks show clear, then you will have very fair weather. Fair weather will be yours too when a low cloud shows upon the broad sea and does not stand high, but is pressed flat there like a ledge of rock. In fair weather look the more for storm, and out of storm for calm. And you must look right well at the Manger, about which the Crab wheels, when first it is cleared of all the mist beneath; for it clears as the storm is waning.

999

Let quiet lamp-flames too, and the owl of the night singing quietly, be your sign that a storm is dying away, and the crow when at the evening hour she caws with many notes, quietly varying them; and the ravens, when alone and solitary they cry twice and then afterwards call in a rush of sound; and when more of them in a flock, as they think of their roost, are full of voice. One would think them glad, such are the cries they give, like clear-voiced callers, and so often about the bark of the tree, or at times upon the tree itself where they roost, do they flap their wings on coming home. And cranes too, it may be, before a gentle calm will stretch out unswerving on a single course, all in one company, and in the fair weather will not be swept back upon their track.

Signs of Storm

1013

But when the clear light from the stars is dulled, and no massed clouds from anywhere come across them, nor does any other darkness run beneath, nor the moon, but they simply, of a sudden, move feebly on, let this no longer stand with you as a sign of calm, but look for a storm; and also when some clouds stay in the same place and others move beneath them, some passing by and some following behind.

1021

Geese hurrying with clamour to their feeding are a great sign of storm, and the crow of nine generations singing by night, and jackdaws crying late, and the chaffinch piping at dawn, and all the birds fleeing in from the open sea, and the wren or the robin slipping into hollow clefts, and the tribes of jackdaws coming from dry pasture to a late roost.

1028

Nor, when a great storm is coming, will the tawny bees range far afield to gather wax, but there at home they circle about their honey and their works; nor do the long lines of cranes on high stretch along the same paths, but wheel about and go back the way they flew. Nor, when in windless air fine cobwebs drift, and the flames of the lamp flicker as they fade, or fire is hard to kindle, and lamps too, in fair weather, be easy about a storm. Why tell you all the signs there are for men? For even from mean ash, caking where it lies, you may infer snowfall, and snow from a lamp, when it has marks like millet seeds all in a ring near the fire-bright wick, and hail from live charcoal, when the coal itself looks bright but in its midst there shows something like a thin cloud, within the burning fire.

Signs of the Seasons: Plants and Animals

1044

Holm-oaks laden with fruit, again, and the dark mastich trees are not untried; everywhere many a farmer is always peering about, afraid that the harvest may slip from his hand. Holm-oaks bearing in due measure a thick-set crop of acorns will tell of a winter that is to grow

strong beyond the common. May they not be weighed down everywhere to wild excess, and may the ploughlands, far from drought, come to ear along with them. Three times the mastich conceives, three are its growths of fruit, and each in turn bears signs for the ploughing. For men divide the ploughing season too in three, the middle and the two ends; the first fruit announces the first ploughing, the middle fruit the middle, and the last of them all the latest. For whichever fruit the teeming mastich bears most finely, the ploughing that answers to it will be richest beyond the others in corn, with the poorest fruit a scanty one, and with the middling a middling. In the same way the stalk of the squill flowers thrice at the top, for marking the signs of a like harvest. All that the ploughman noted in the fruit of the mastich he reads also in the white flower of the squill.

1064

But when in autumn wasps in great multitudes swarm thick everywhere, one may tell even before the Pleiades are in the west[21] that winter is coming on, such is the whirl that all at once eddies among the wasps. And when sows and ewes and she-goats turn back to mating, and after receiving all the males are mounted yet again a second time, they, just like the wasps, will tell of a great winter. But when goats and sheep and swine mate late, the poor man is glad, because their coupling shows him, who is none too warm, a mild year. The timely ploughman is glad too of flocks of cranes that come in season, and the untimely one in his turn rather of the late[22]; for the winters come on just as the cranes do: early when they come early and more in companies; but when they appear late and not in flocks, and pass over a longer time and not many together, the later works are the better for winter's delay.

1082

If oxen and sheep dig the ground after laden autumn and stretch their heads against the wind Boreas, then the Pleiades themselves at their setting will bring a very stormy winter. May they not dig too much, for then the winter is great beyond due measure, and a friend neither

to trees nor to tilled fields. Rather let there be deep snow on the great ploughlands, over the blade not yet parted nor grown tall, so that the man who waits may rejoice in well-being. And may the stars above be always like themselves, and none long-haired[23], not one nor two nor more; for many are long-haired in a year of drought.

1094

Nor is the man of the mainland glad of flocks of birds, when in multitudes from the islands they descend upon his fields as summer is coming on; he is terribly afraid for his harvest, that it may come to him empty and all chaff, afflicted by drought. But the goatherd, it may be, is glad of the very same birds, when they come in due measure, looking forward to a year thereafter rich in milk. For so we men live, toiling and wandering, one by one means and another by another. But all are ready to recognise the signs that lie at their feet and to make them their own for the moment.

Signs of Storm from Animals

1104

By their sheep shepherds judge of storms, when they run to pasture more eagerly than usual, and some out of the flock, rams here and lambs there, play along the way, butting with their horns; or when here and there they kick up their feet, the nimble ones with all four and the horned with two; or again when the men move them unwilling from the flock[24], driving them in at evening all the same, and they nibble at the grass on every side though urged on with showers of stones.

1113

From oxen ploughmen and herdsmen learn of a storm stirring; for when oxen lick round the hooves of the forefoot with their tongue, or stretch themselves upon their right sides in their lair, the old ploughman looks for a putting off of the ploughing. Nor, when the cows gather full of lowing as they come to the steading at the hour of ox-loosing, do the heifers, sullen at leaving the meadow and the

cattle-pasture, give sign that they will presently feed their fill without storm. Nor are goats eager about the prickles of the holm-oak a sign of fair weather, nor sows raving in the rubbish.

1124

And when a wolf, a lone wolf, howls long, or when, little wary of ploughmen, he comes down to the tilled lands like one in want of shelter, close to men, to have his bed there, look for a storm when dawn has come round three times. So too by the earlier signs you may judge of winds to be, or storm, or rain, on the day itself, or the one after, or even on the third dawn.

1132

Nor indeed did mice, if ever in fair weather they squeaked more than usual and skipped as though in dances, go unobserved by the men of older time; nor dogs; for the dog digs with both paws when he expects a storm coming on, and those mice too are then prophets of storm. And the crab too goes from the water to dry land when a storm is about to rush upon its way. And mice by day, tossing their bedding with their feet, long for sleep when signs of rain are showing.

Conclusion

1142

Scorn none of these. It is good to look for sign upon sign; when two go the same way there is more ground for expectation, and with a third you may be confident. Always count up the signs as the year goes by, comparing whether at the rising or the setting of a star there dawns such a day as the sign foretold. It will be of great service to mark the two fourth days, of the waning month and of the month beginning; for they hold the bounds where the months meet, when for eight nights the sky is more treacherous, for want of the pale-gleaming moon. If you have watched all these together through the year, you will never judge of the sky off-hand.

Notes

[1] The "earlier generation" (προτέρη γενεή) is left dark by Aratus: it has been taken as the elder gods, the heroes of old, or the first observers of the stars.

[2] Aratus derives the name Wains (Ἅμαξαι) from ἅμα, "together".

[3] The Kneeler is the constellation later identified with Heracles (Hercules); Aratus leaves it nameless and calls it simply "the Figure" at 73 and 270.

[4] The Ear of Corn is the star Spica.

[5] The Vintage-Herald (Προτρυγητήρ) is Vindemiatrix, ε Virginis; lines 138 and 142a are bracketed by Mair as interpolations and are translated as printed. From 139 to 148 the comparison carries the poem over to the Great Bear, whose own stars and neighbours are described.

[6] Zeus here stands for the sky itself, as often in the poem: so "in father Zeus" (253), "out of Zeus" (259), "runs along with Zeus" (275) and "when Zeus is fair" (899).

[7] The Knot is the star α Piscium.

[8] The Bull is figured as a forepart only, cut off at the middle.

[9] This ring of stars under the Archer's forefeet, left unnamed by Aratus, is the Southern Crown.

[10] That is, when the Centaur's shoulder stands on the meridian, midway between its rising and its setting.

[11] The ray from the eye to the sky is the radius of the circle; six lines of that length span the circle as an inscribed hexagon, each cutting off two of the twelve signs.

[12] The horns of Ocean are the eastern and western horizons; so too "the other horn" at 589.

[13] The Fish here is the Southern Fish; "he who is turned backward" (575) is the Kneeler.

[14] Ox-loosing (βουλυτός), the evening hour of unyoking, plays on the name Boötes, the ox-driver, who takes more than half the night to set.

[15] The printed μειρομένη γονάτων is translated as it stands; the phrase is difficult and has been variously emended.

[16] The nineteen cycles of the sun are the nineteen-year cycle of Meton, which reconciles the lunar months with the solar year.

[17] Parhelia are mock suns, bright patches seen on either side of the sun.

[18] The Manger is the misty cluster Praesepe in the Crab; the Asses are the two faint stars north and south of it, γ and δ Cancri.

[19] The ὀλολυγών is of uncertain identity: it is usually taken as a tree-frog, sometimes as a bird.

[20] Translated as printed (χείματος ... ὑπέτυψε); other editors read κύματος ... ὑπέκυψε, "ducks under the wave as it comes to land".

[21] The Pleiades setting in the west at dawn, in early November, marked the onset of winter.

[22] The Greek is compressed, literally "and the untimely one in turn rather": the late ploughman is glad when the cranes, and so the winter, come late.

[23] The long-haired stars are comets.

[24] The printed ἀγλης is a misprint for ἀγέλης, "flock", which is translated.